Odds & Ends: A Real Collaboration

Mixed Models in Practice — Simulated Solar Radiation and Human Nevi

Emile Latour

March 18, 2026

The collaboration

I have a long-term working relationship with the **OHSU Department of Dermatology**.

This project came out of that relationship:

The scientific question

Does exposure to **simulated solar radiation (SSR)** change the expression of specific RNA markers in human melanocytic nevi (moles)?

Nevi are benign melanocyte clusters — and a key focus of melanoma prevention research.

My role: **statistical collaborator**. I didn't design the experiment or collect the tissue — I was brought in to figure out how to analyze it correctly.

Why this matters to you

This is what collaboration looks like in practice.

The researchers had a biologically important question and a carefully designed experiment. They needed someone who could:

- Understand the data structure
- Choose the right model
- Communicate the results clearly

That's the role this course prepares you to play — or to seek out.

The study design

6 volunteers, each contributing **2 nevi**:

- One nevus received SSR (the irradiated condition)
- One nevus served as an unirradiated control
- Both nevi from the same person were matched for size, location, and clinical appearance

The tissue was excised by punch biopsy and analyzed using **single-nucleus RNA sequencing (snRNAseq)** — a technique that measures gene expression at the level of individual cell nuclei.

Three RNA markers were of primary interest:

- **SERPINE2** — a melanoma-associated serine protease inhibitor
- **CCL21** — a marker of lymphatic endothelial activation
- **IGFBP4** — a modulator of the IGF-1 signaling axis

Spot the design

Does this remind you of anything from the course?

Each patient contributes **one irradiated nevus** and **one control nevus**.

The patient is their own control. The natural comparison is **within-person** — exactly the logic of the **paired t-test**.

The data structure problem

Here's where the paired t-test breaks down — and where a new tool is needed.

A paired t-test requires **one summary value per person per condition** — a single difference per pair.

But snRNAseq doesn't give you one number per sample. It gives you **hundreds of individual cell-nucleus measurements** per sample.

For each patient, for each RNA, for each condition, we have a **distribution of expression values** — one per nucleus captured.

The data structure

```
ptid  | condition  | RNA      | value
-----|-----------|---------|------
001   | Control    | SERPINE2| 12.3
001   | Control    | SERPINE2| 8.7
001   | Control    | SERPINE2| 15.1
...   | ...        | ...     | ...
001   | SSR        | SERPINE2| 24.6
001   | SSR        | SERPINE2| 19.2
...
002   | Control    | SERPINE2| 10.4
...
```

Hundreds of rows per patient per condition. All observations within a patient are **correlated**.

The model

The solution is the **linear mixed model** — exactly what we introduced in the animal studies section. The key idea: give each patient **their own baseline**.

In R:

```r
library(lme4)

model <- lmer(
  value ~ tx + (1 | ptid),
  data = rna_data
)
```

What each piece does

- `tx` — fixed effect for treatment (SSR vs. Control): the effect we care about
- `(1 | ptid)` — random intercept for patient: each person gets their own baseline, absorbing the within-patient correlation
- Estimates the treatment effect **within patients**, not just across group averages

Connection to the paired t-test

The random intercept does conceptually what the paired t-test does — it removes between-person variability so the treatment effect comes from **within-person differences**.

The mixed model generalizes this to settings where you have many measurements per person, not just one.

The model

The solution is the **linear mixed model** — exactly what we introduced in the animal studies section.

$$\text{Expression}_{ij} = \beta_0 + \beta_1 \cdot \text{SSR}_i + b_j + \varepsilon_{ij}$$

In R:

```r
library(lme4)

model <- lmer(
  value ~ tx + (1 | ptid),
  data = rna_data
)
```

What each piece does

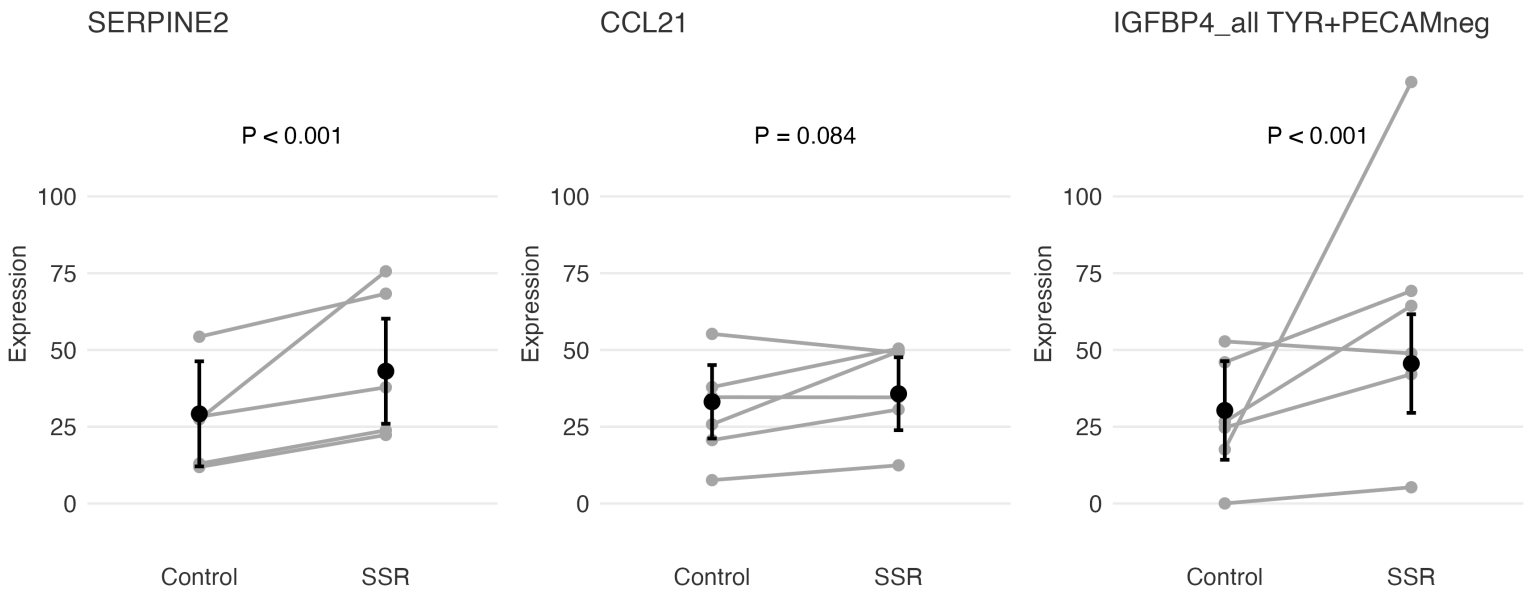
- `tx` — fixed effect for treatment (SSR vs. Control): the effect we care about
- `(1 | ptid)` — random intercept for patient: each person gets their own baseline, absorbing the within-patient correlation
- This estimates the treatment effect **within patients**, not just across group averages

Connection to the paired t-test

The random intercept for patient does conceptually what the paired t-test does — it removes between-person variability so the treatment effect is estimated from **within-person differences**.

The mixed model generalizes this to settings where you have many measurements per person, not just one.

The results



What to look for

Each gray line is one patient. Black points and error bars are model-estimated marginal means ± 95% CI. p-values from the linear mixed model.

The gray lines tell the within-patient story. When most lines slope in the **same direction**, the within-patient signal is strong — and that's what the mixed model is detecting. It's possible for the group-level confidence intervals to **overlap** while the within-patient effect is still statistically significant, because the model uses the paired structure directly. Notice also that CCL21 shows a trend (p = 0.084) but not a significant result — not every marker responds the same way, and the model reflects that honestly.

What I learned from this collaboration

Statistically:

- The biology determined the model. The paired design was built in — my job was to recognize it in the data structure and use a model that respected it.
- Mean-centering within patients (a sensitivity analysis) confirmed the results weren't driven by between-patient baseline differences. The findings held.
- Communicating "why the group CIs can overlap but the test is still significant" to non-statisticians is one of the hardest and most important things we do.

About collaboration:

Tip

The researchers knew their biology deeply. They knew what nevi are, what SSR does to DNA, why these three RNA markers matter for melanoma prevention.

I knew the data structure and what model to use.

Neither of us could have done the other's job. That's what makes it a real collaboration — and why the conversations you'll have with statisticians matter. **You bring the biology. We bring the model.**